
Decoding Disease, Designing Interventions

Harnessing AI to Bridge Disease Understanding and Therapeutic Design

ICS Honors Seminar



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Information & Computer Sciences

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The Data-to-Intervention Gap

Biological Data Are Growing. Treatments Are Still Limited.

The need

1 in 4 adults
experienced a mental illness



1 in 10 people
had a drug use disorder



U.S., 2024

The data

We can measure more than ever



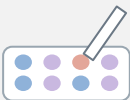
Gene evidence

genetics, literature,
and databases



Cell profiles

millions of cells
× 20,000 genes



Perturbation data

what happens when
genes change

The gap

~90% fail
of drug candidates in clinical trials

Data alone don't tell us:

- 1 How are cells organized?
- 2 How do cells interact?
- 3 Which genes should we target?
- 4 Which gene combinations could restore a healthy cell state?

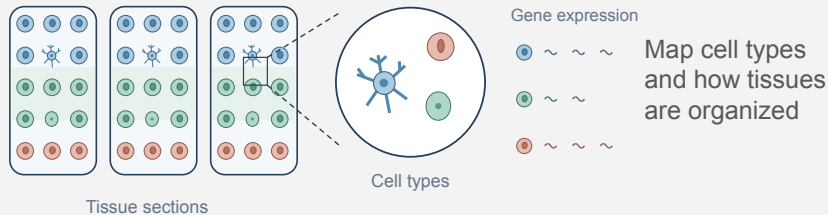
Closing this gap is what our lab works on.

Rounded estimates. 2024 U.S. NSDUH. Drug use disorder: 9.8%, ages 12+. Any mental illness: 23.4%, ages 18+.

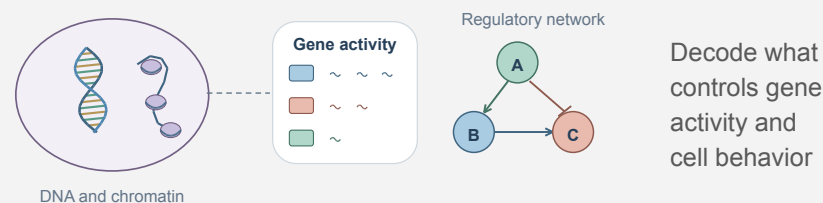
What Does the Zhang Lab Study?

Using AI to understand biological systems and explore interventions

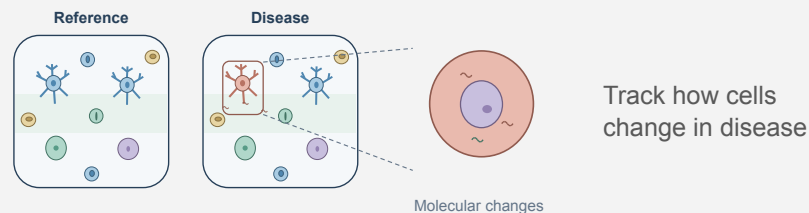
Single-Cell & Spatial Genomics



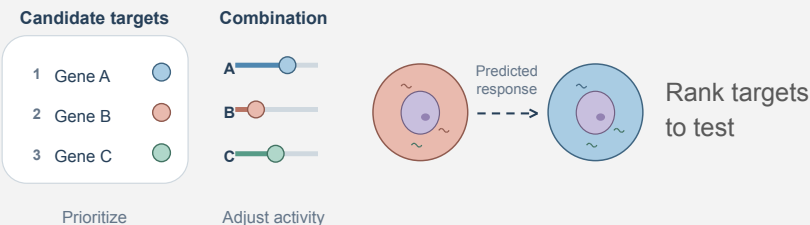
Gene Regulation & Functional Genomics



Human Disease Genomics

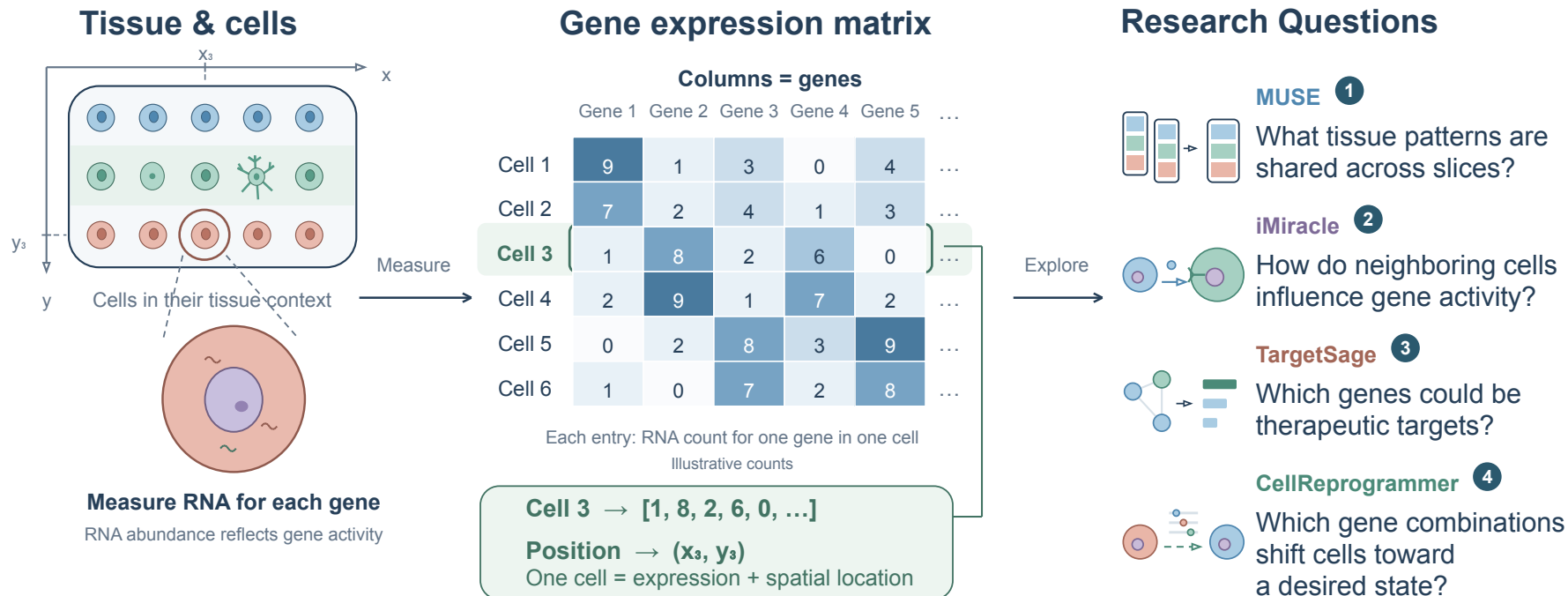


AI-Guided Therapeutic Discovery



From Cells to Data

Measuring gene activity gives each cell a molecular profile



01

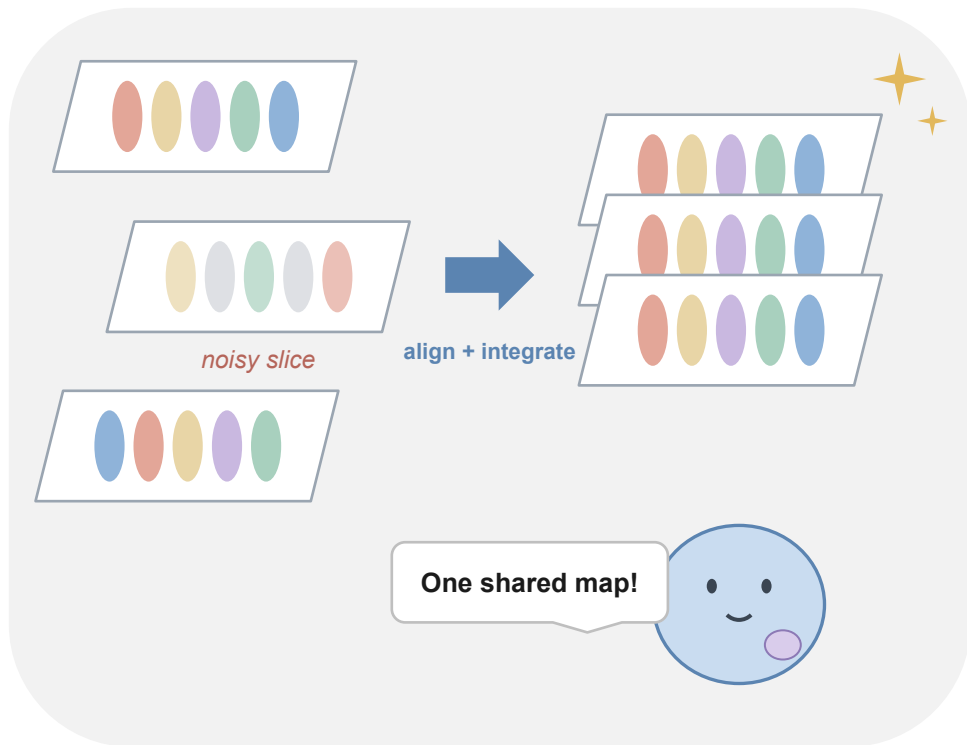
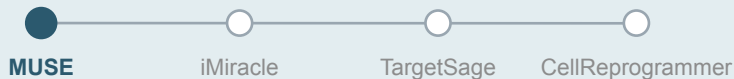
DECODING DISEASE

MUSE

CIKM 2025

What tissue patterns are
shared across slices?

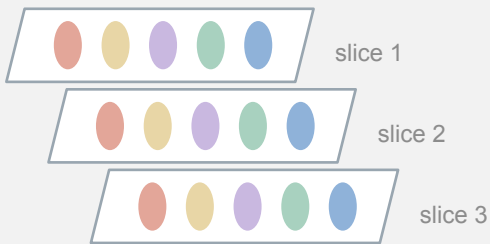
*Adjacent slices share structure,
so cleaner slices can help noisier ones.*



Multi-Slice Integration: Alignment and Data Quality

Misalignment and noise obscure shared tissue patterns.

One Tissue, Many Slices



A tissue is cut into thin slices.
Each spot has an (x, y) location
plus counts for ~20,000 genes.

Slices Don't Match Up



Each slice is measured on its own,
so the same tissue region can
look different from slice to slice.

Some Slices Are Noisy



Low-quality slices have missing or
noisy measurements that hide the
real patterns.

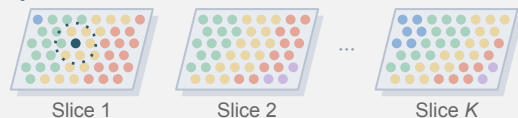
MUSE's goal: match spots across slices and let clean slices help noisy ones → one shared tissue map

MUSE: Cross-Slice Alignment and Quality-Aware Integration

Three steps that upgrade existing single-slice methods to many slices

1 Build Spatial Graphs

Input: multi-slice data

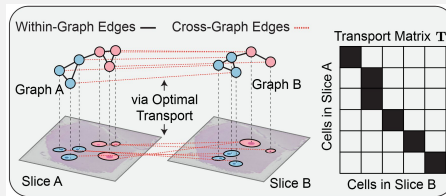


Spatial graph per slice



Link each spot to its nearest neighbors, so every spot knows its local context.

2 Match Across Slices

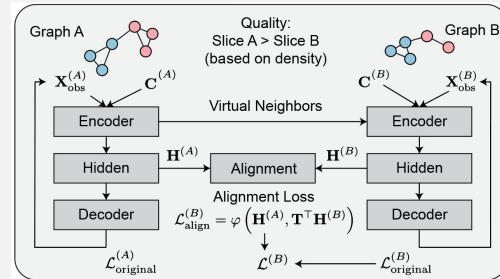


$$\min_{T \geq 0} (1 - \alpha) \underbrace{\sum_{i,j'} \delta(\mathbf{x}_i^{(A)}, \mathbf{x}_{j'}^{(B)}) T_{ij'}}_{\text{gene expression similarity}} + \alpha \underbrace{\sum_{i,p,j',q'} (D_{ip}^{(A)} - D_{j'q'}^{(B)})^2 T_{ij'} T_{pq'}}_{\text{spatial context preservation}}$$

Alignment criteria:

- similar expression profiles
- similar local neighborhoods

3 Quality-Aware Fusion

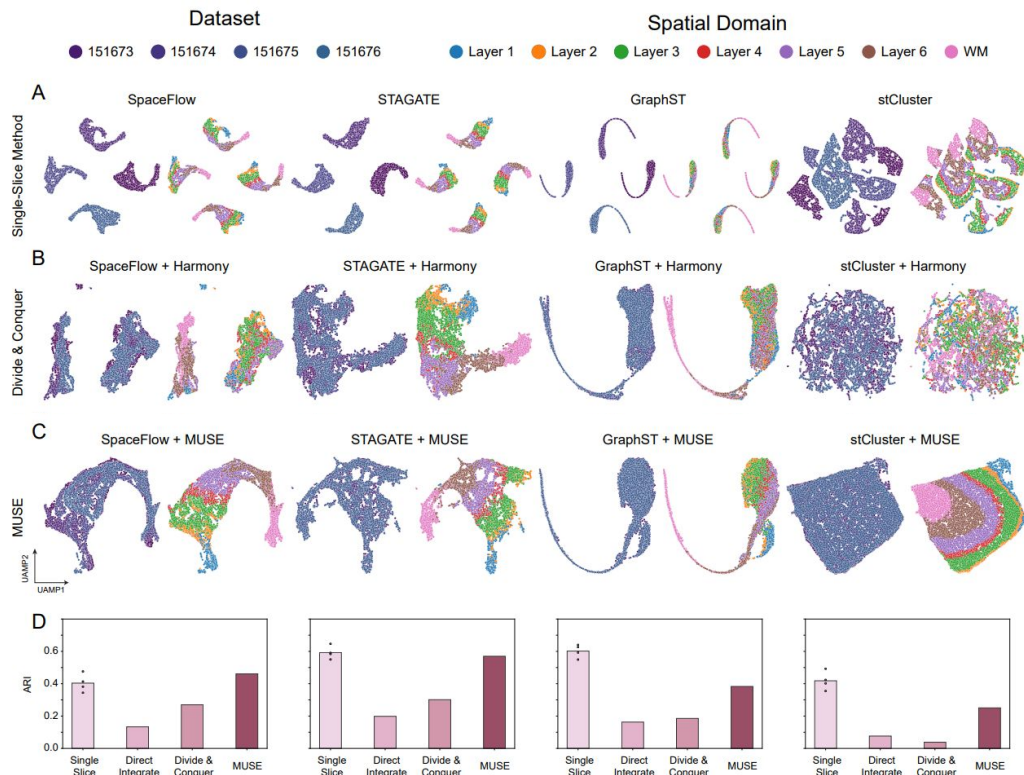


Virtual neighbors:

$$i = \arg \max_i T_{ij'}, \quad N_{j'} = N_{j'}^{\text{original}} \cup \{i\}$$

Noisy slices borrow “virtual neighbors” from a reference slice.

MUSE Learns Coherent Embeddings Across Slices



Human DLPFC (10X Visium)

4 adjacent cortical slices

Single-slice methods:

Fragmented by dataset

Post-hoc correction:

Slices still form separate groups

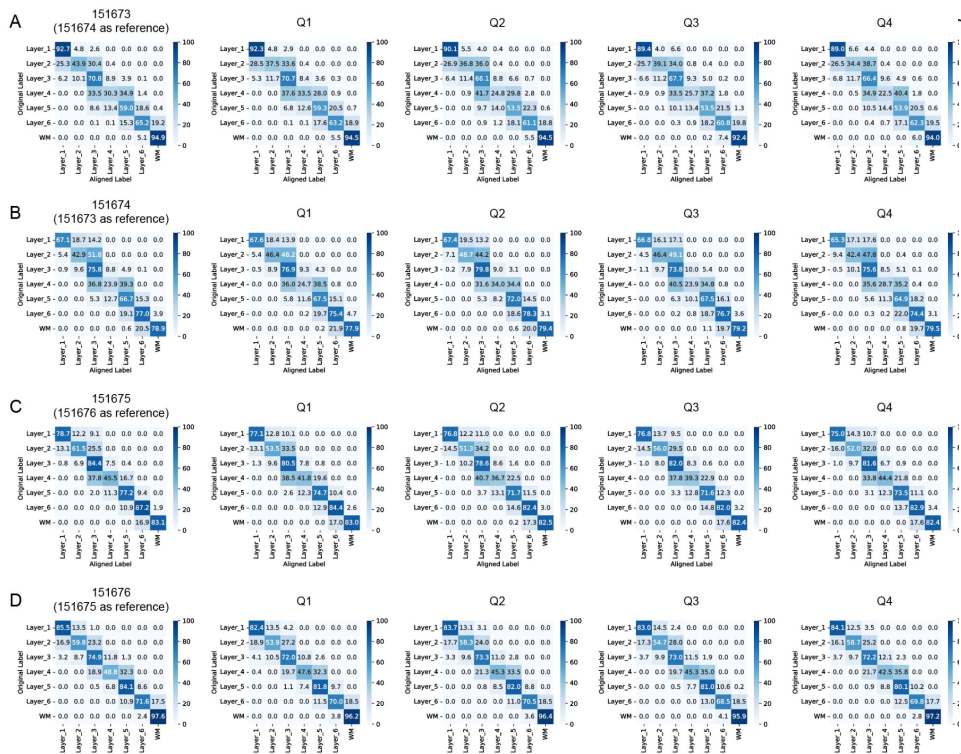
MUSE integration:

Aligned across slices

ARI comparison:

Improved domain recovery

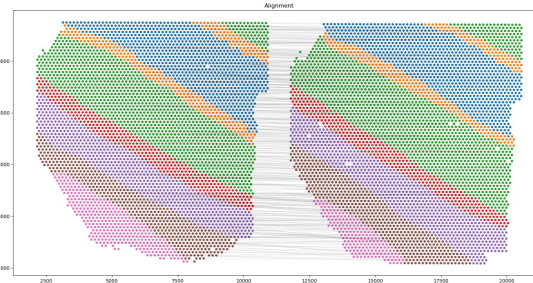
MUSE Aligns Corresponding Cells Across Slices



Strong diagonal signals
accurate layer-to-layer matching

Robust across quality levels
stable from Q1 to Q4

Consistent across references
reproducible alignment patterns



MUSE can accurately align
cells across slices!

02

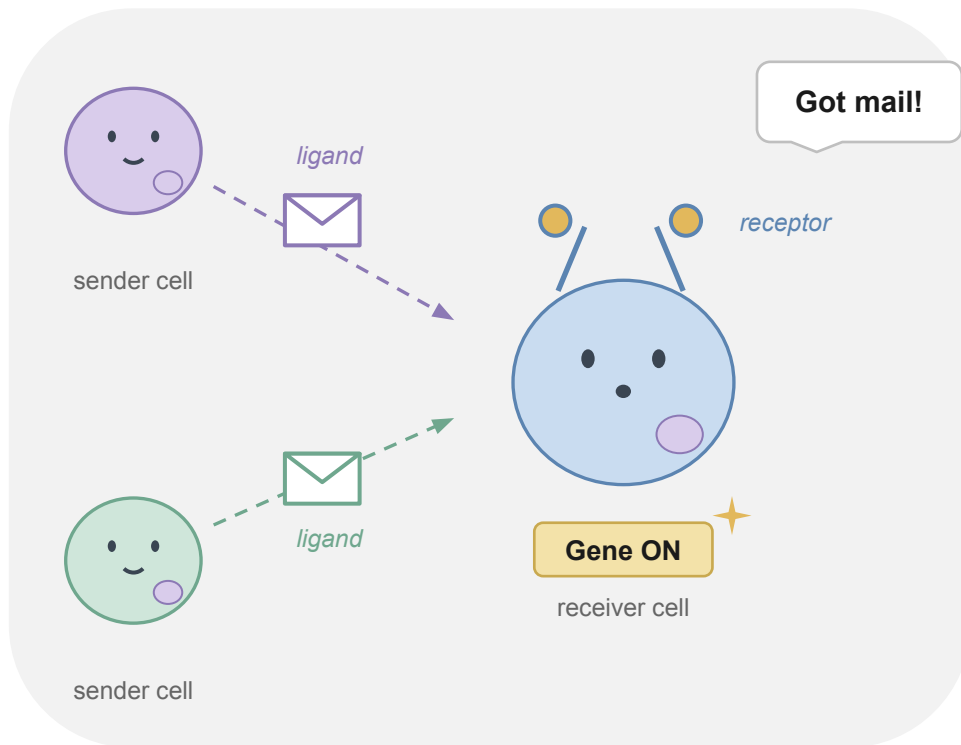
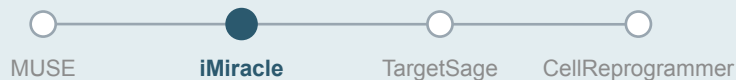
DECODING DISEASE

iMiracle

CIKM 2024

How do neighboring cells influence gene activity?

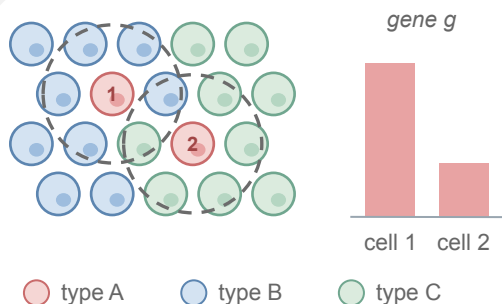
A cell's gene activity combines its own baseline with signals from its neighbors.



Linking Cell–Cell Signals to Gene Expression

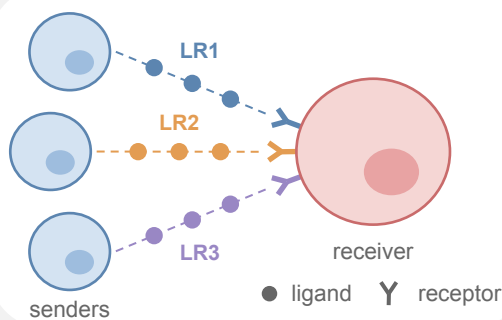
Which signals contribute to each gene's expression in each cell?

Tissue context



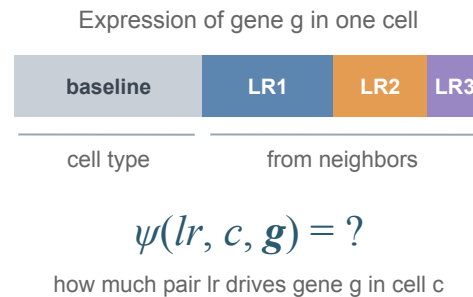
Same cell type, different neighbors
→ different gene activity.

Cell–cell communication



Ligands from many senders bind
receptors on one receiver.

Regulatory interpretation



Which signal explains how much of
each gene, in each cell?

Most existing methods: which cells communicate → **iMiracle:** what those signals do to each gene, in each cell

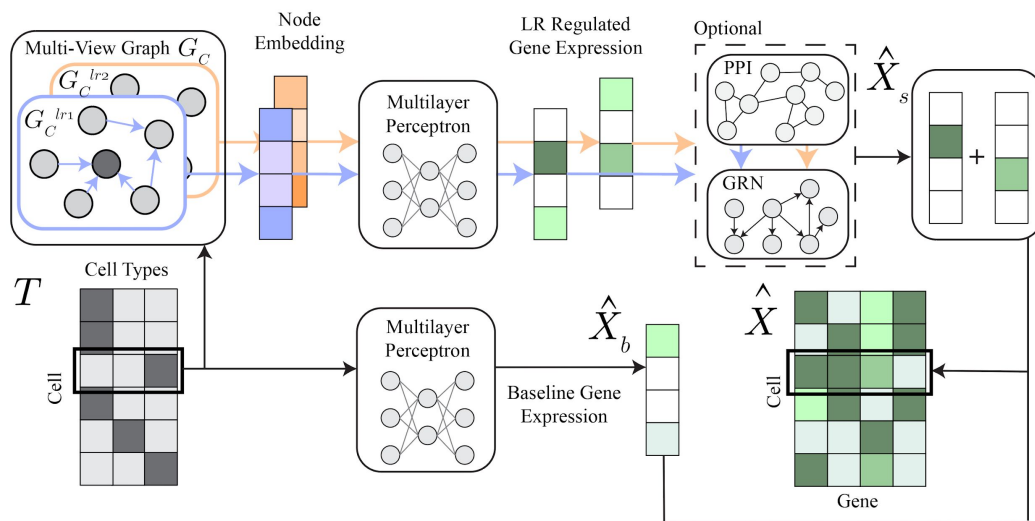
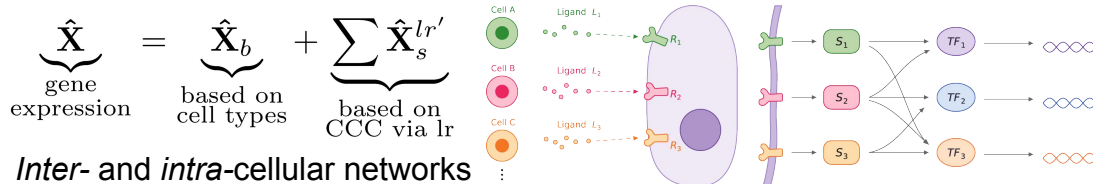
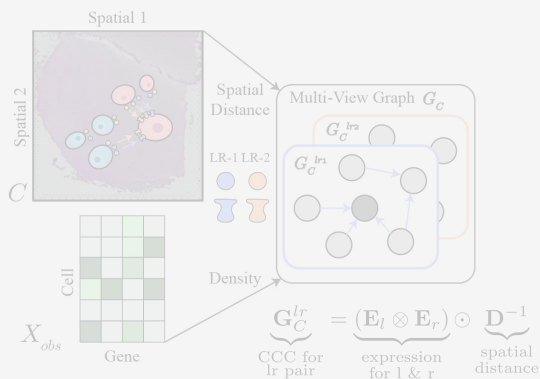
iMiracle: Baseline + Neighbor-Driven Regulation

Multi-View Graph Construction

Ligands: **Molecules** that bind to receptors to trigger a cellular response



Receptors: **Proteins** on cell surfaces that receive signals from ligands



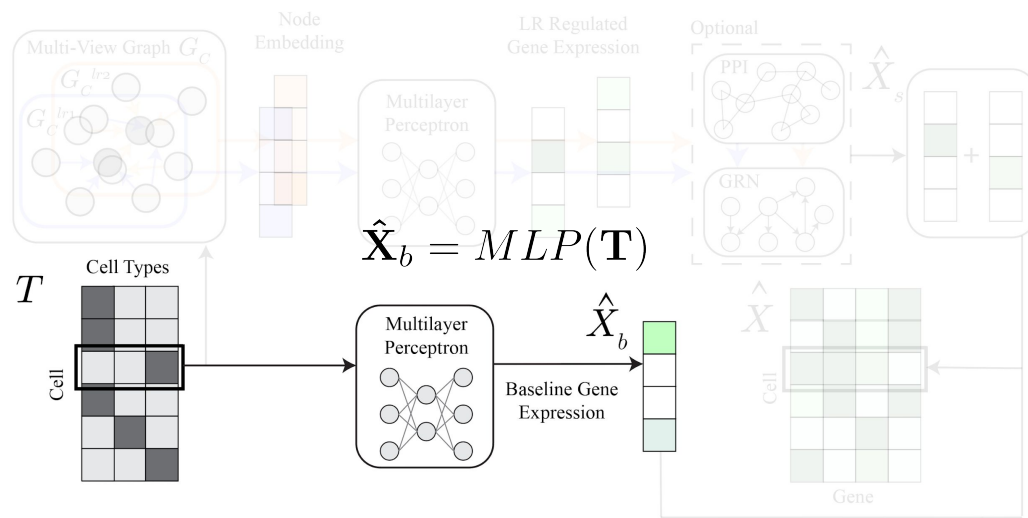
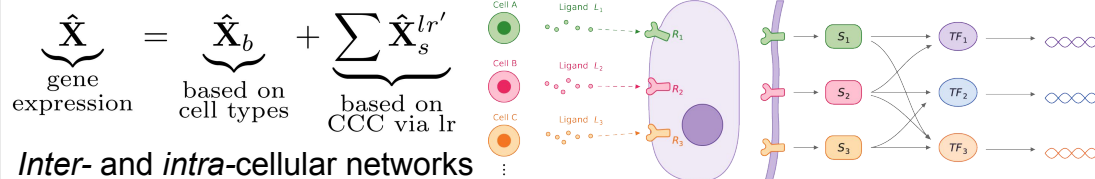
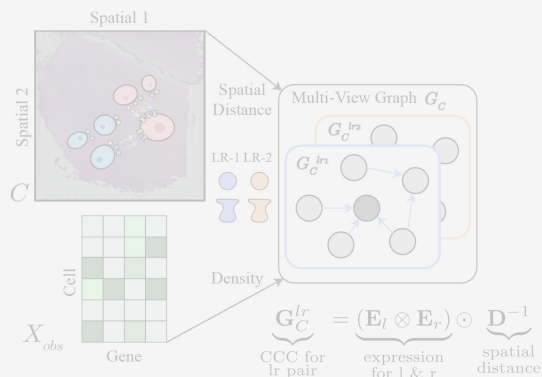
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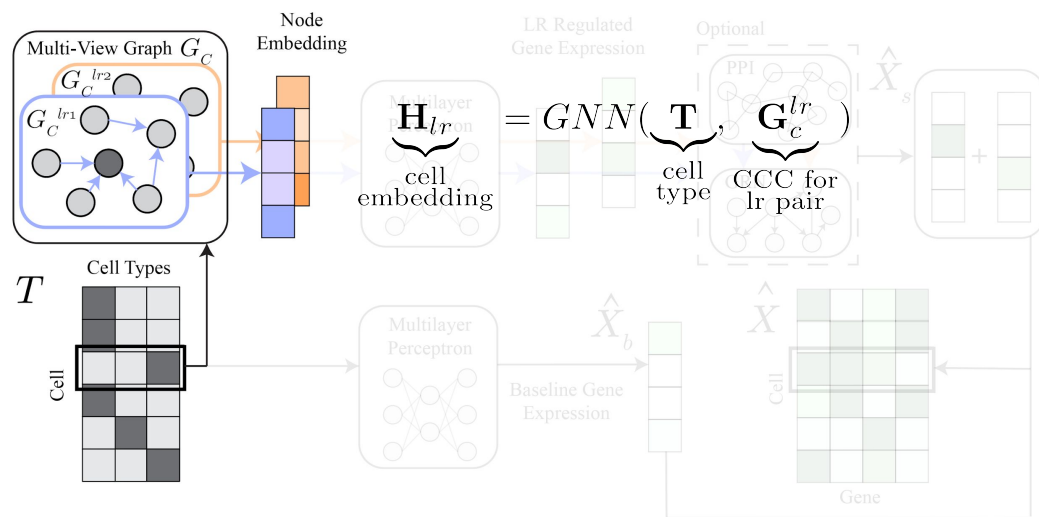
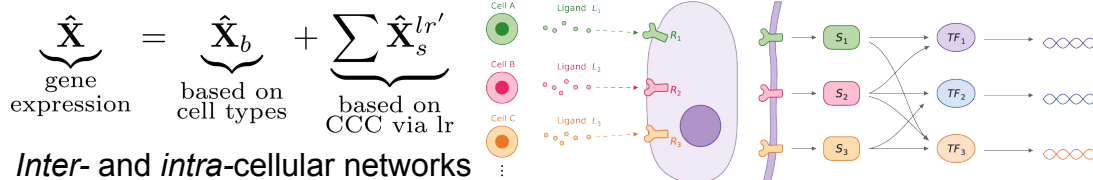
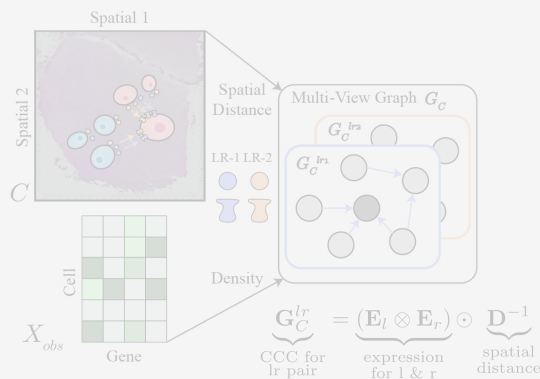
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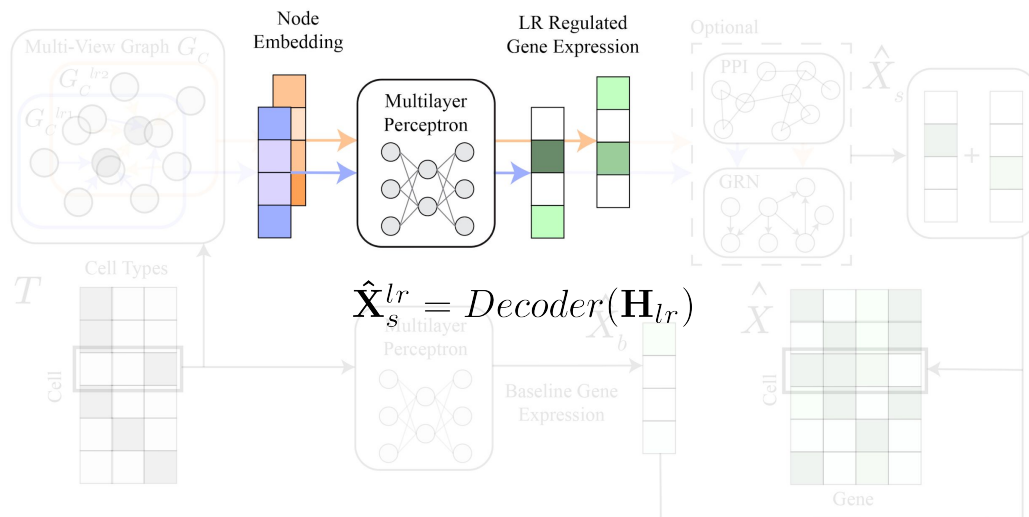
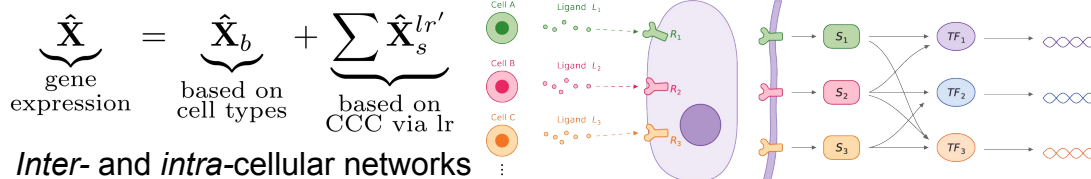
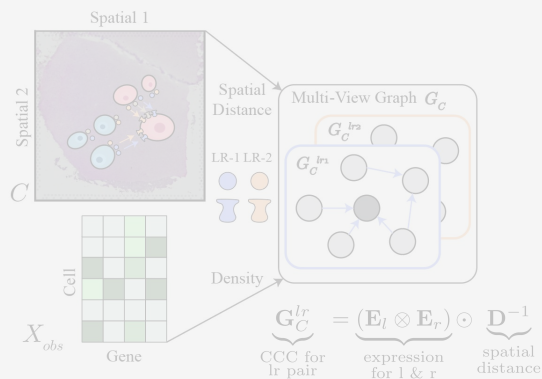
iMiracle: Baseline + Neighbor-Driven Regulation

Multi-View Graph Construction

Ligands: **Molecules** that bind to receptors to trigger a cellular response



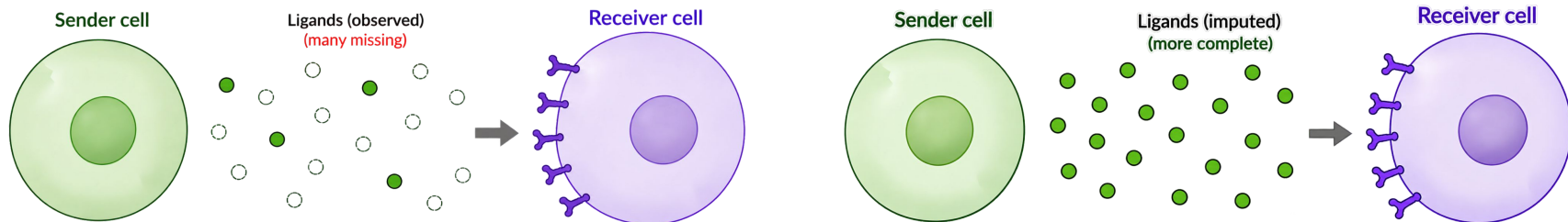
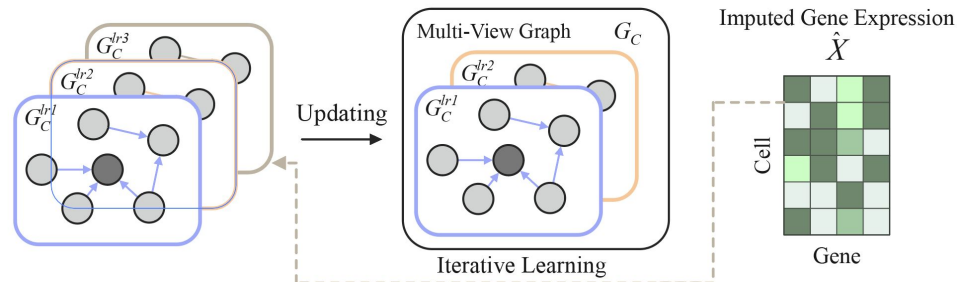
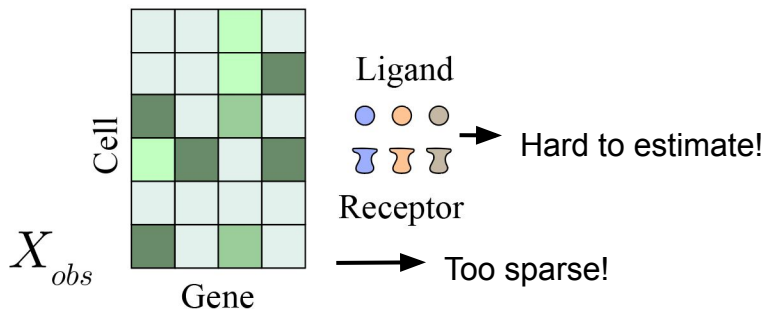
Receptors: **Proteins** on cell surfaces that receive signals from ligands



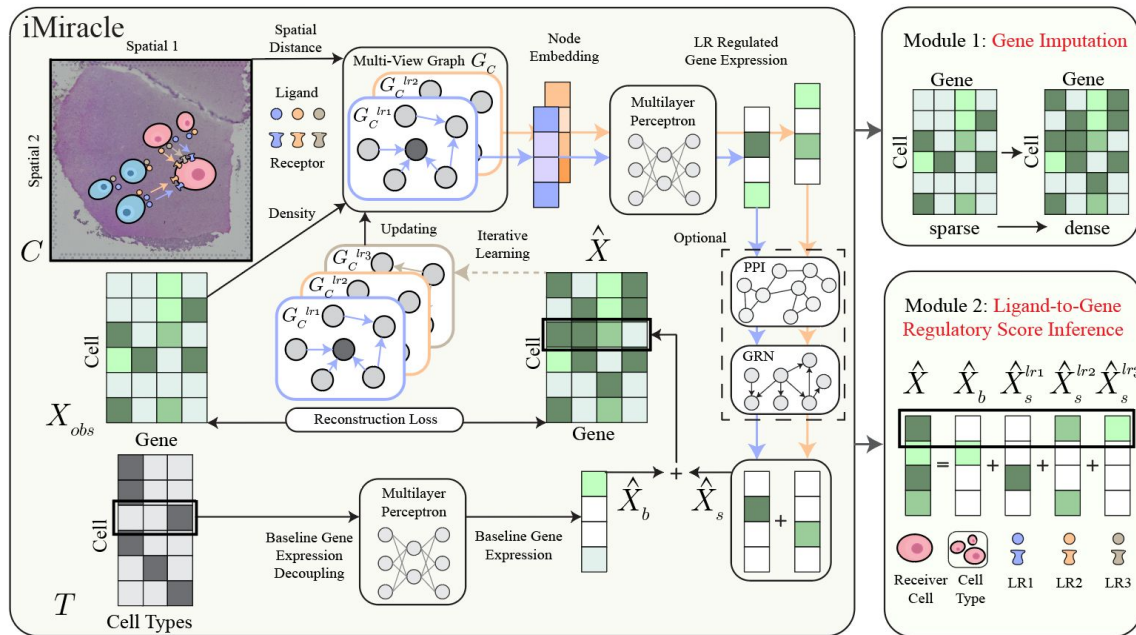
iMiracle: Iterative LR Density Estimation

$$\underbrace{\mathbf{G}_C^{lr}}_{\text{CCC for lr pair}} = \underbrace{(\mathbf{E}_l \otimes \mathbf{E}_r)}_{\text{expression for l \& r}} \odot \underbrace{\mathbf{D}^{-1}}_{\text{spatial distance}}$$

$$\mathbf{G}_c^{lr(i+1)} = \alpha \mathbf{G}_c^{lr(i)} + (1 - \alpha) \mathbf{G}_c^{lr(i+1)}$$



Method Overview



Input

Cell-by-gene matrix

$$X_{obs} \in \mathbb{R}^{n \times m}$$

Cell positions

$$C \in \mathbb{R}^{n \times 2}$$

Cell-by-cell-type matrix

$$T \in \mathbb{R}^{n \times t}$$

Output

Imputed cell-by-gene matrix

$$\hat{X} \in \mathbb{R}^{n \times m}$$

Cell-specific regulatory score

$$\psi(lr, c, g) = \hat{X}_s^{lr'}[c, g] \in \mathbb{R}$$

Model training

$$\mathcal{L} = \frac{\sum_{i,j} \mathbf{1}[X_{obs,(i,j)} \neq 0] (X_{(i,j)} - X_{obs,(i,j)})^2}{\sum_{i,j} \mathbf{1}[X_{obs,(i,j)} \neq 0]}$$

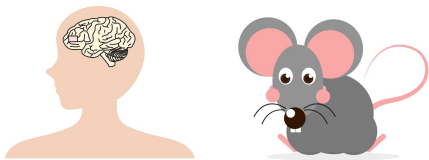
Reconstruction error over observed (non-zero) entries, so no ground-truth regulation is needed.

iMiracle Results: CCC Recovery & Gene Imputation

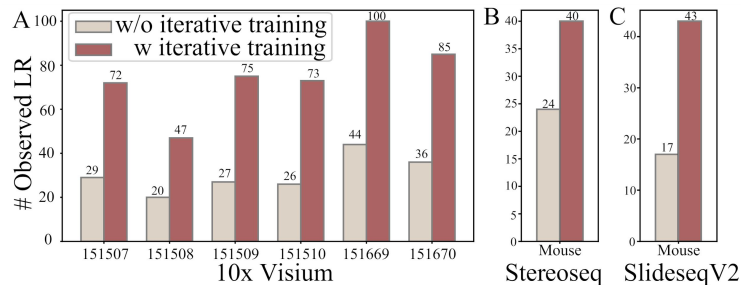
Datasets

2 species

- dorsolateral prefrontal cortex
- mouse olfactory bulb

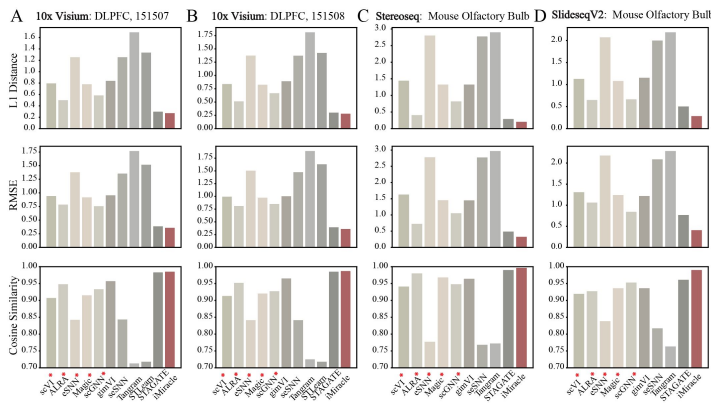


iMiracle fully delineates the CCC network via iterative learning



iMiracle improves the number of detected LR interactions.

iMiracle shows improved gene imputation accuracy



We can confidently rely on imputed gene expression to estimate LR density!

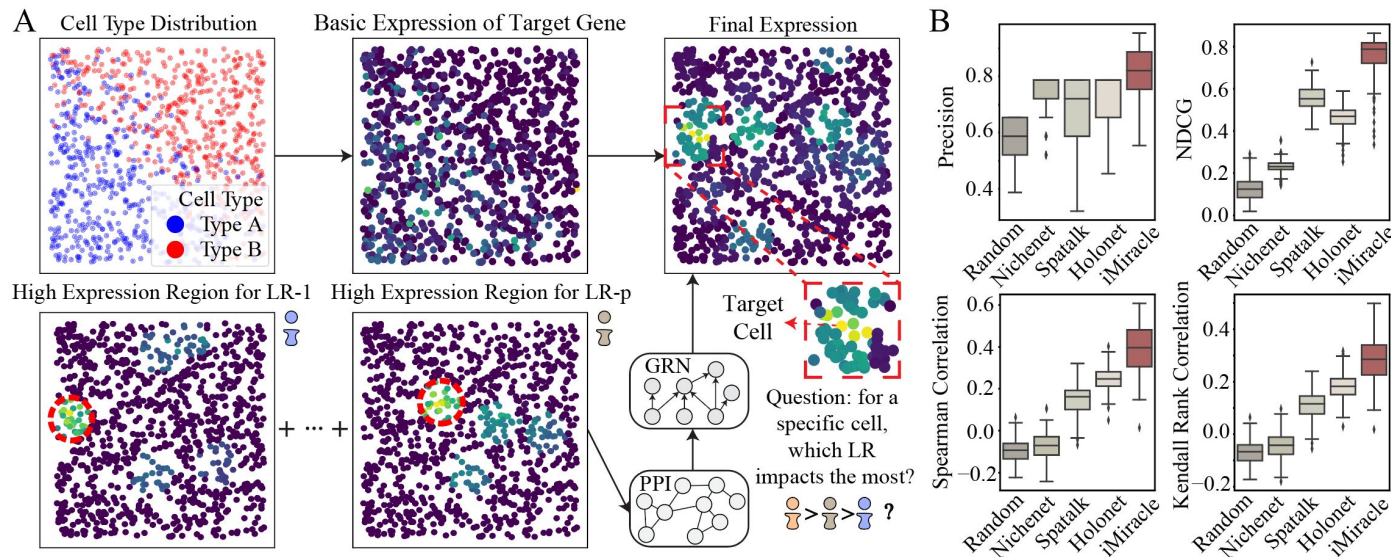
3 sequencing platforms

- 10X Visium
- Stereo-seq
- Slide-seqV2

8 datasets

iMiracle Results: LR–Gene Regulation

iMiracle infers accurate inter-cellular ligand-gene regulatory scores



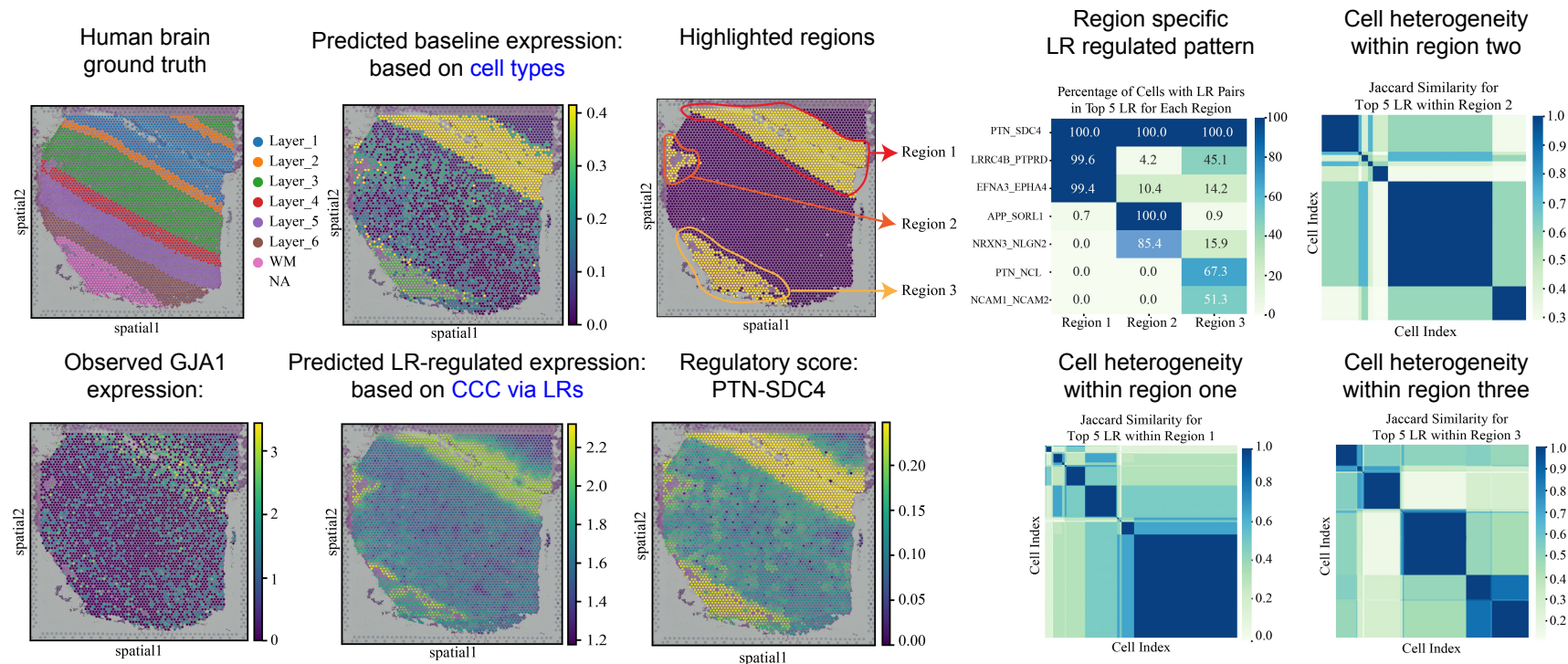
Simulated datasets:

We know the ground truth, specifically which LR contributes the most to a gene in a given cell.

iMiracle is **more accurate** than random guess and methods with/without spatial information!

iMiracle Results: Regulatory Heterogeneity

iMiracle reveals substantial regulatory heterogeneity across cells



03

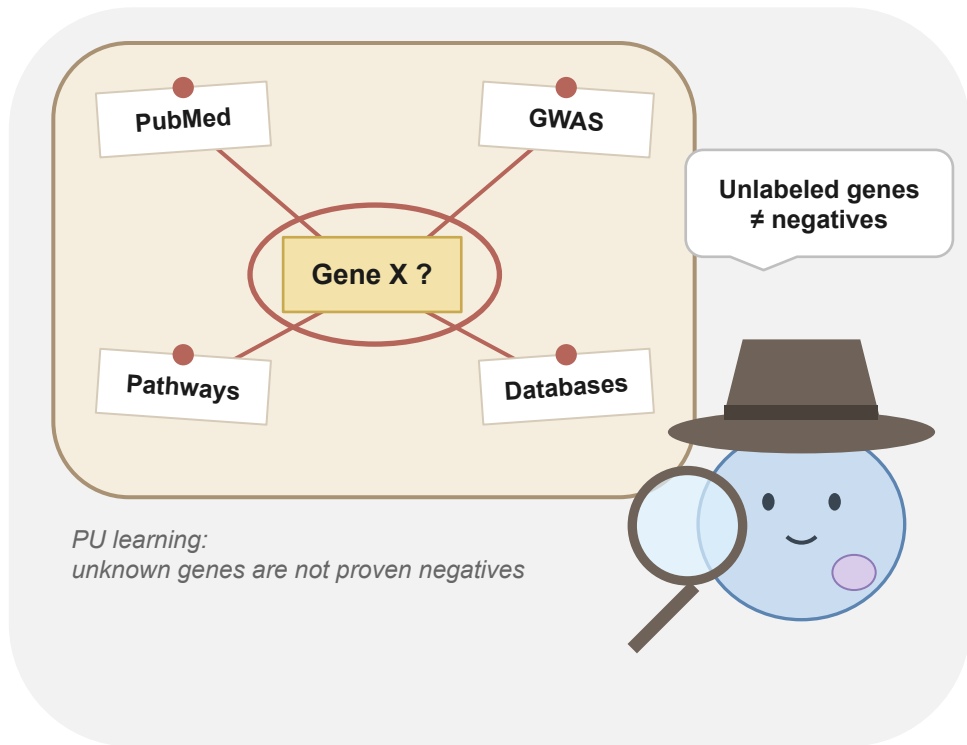
DESIGNING INTERVENTIONS

TargetSage

NeurIPS 2026

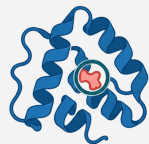
Which genes could be therapeutic targets?

Evidence is scattered across sources, and only a few hundred genes are known targets.

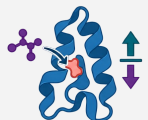


Target Identification: Challenges and Our Approach

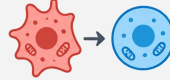
What are we looking for?



Identify a target



Modulate activity

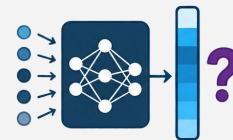


Therapeutic benefit

Three challenges



Static features



Limited interpretability



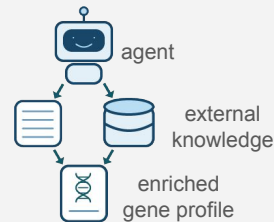
Incomplete labels

How do existing methods rank candidates?

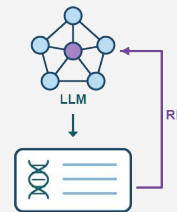


DrugnomeAI: 324 gene-level features from 15 data sources

How does TargetSage address them?



Agentic Profiling



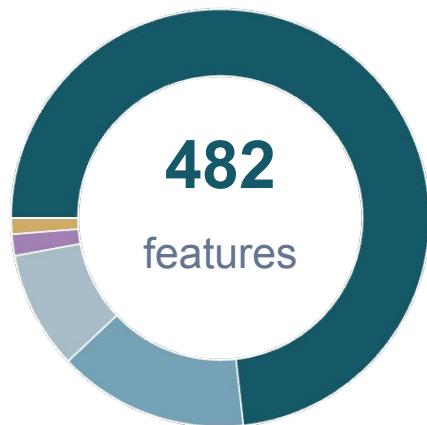
Guided Reasoning



TargetSage: Integrating Evidence to Prioritize Targets

Biological evidence

19,032 human protein-coding genes



- **Function & networks (353)**
- Chemical & antibody links (71)
- Genomic constraint (44)
- Phenotype & disease (8)
- Expression & sequence (6)

+ Retrieved biomedical evidence (literature & databases)

Training examples

700 known drug targets

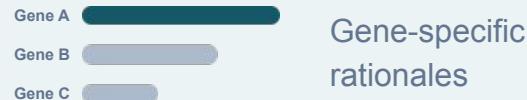
18,332 unlabeled genes

PHAROS: a database of drug targets



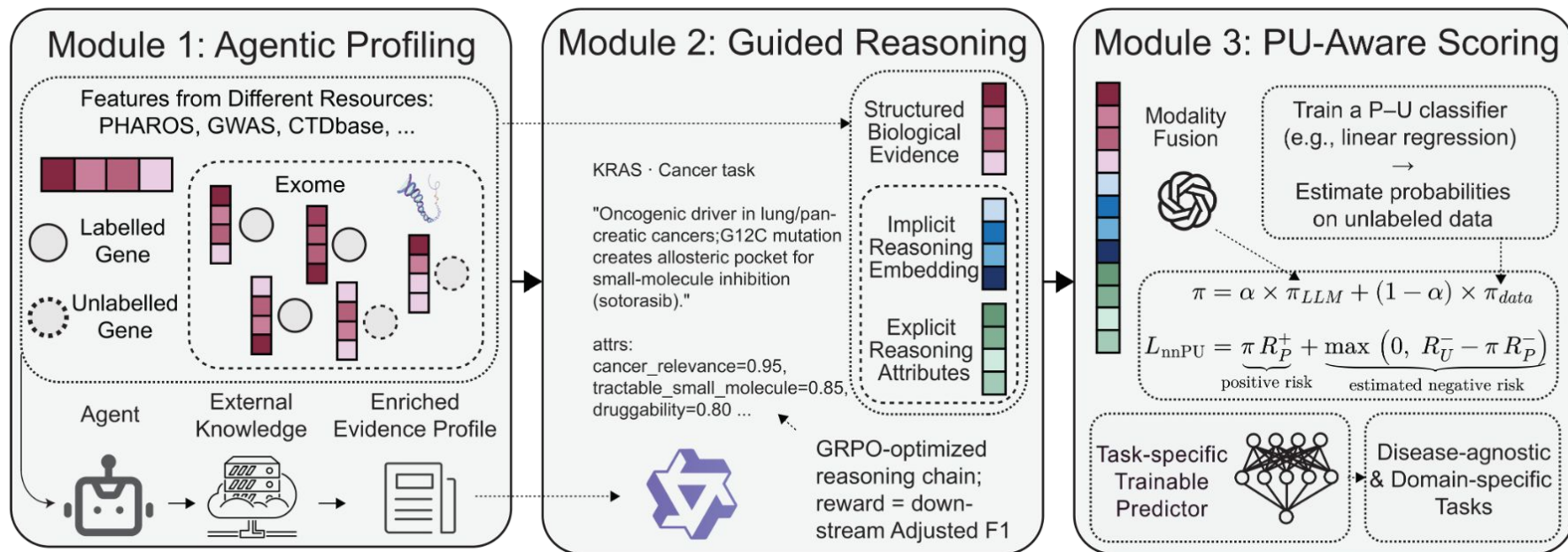
TargetSage

Target prioritization



TargetSage Turns Evidence Into Target Scores

INPUT: Gene evidence + known targets → OUTPUT: Target scores + rationales



482 features · 15 databases
6-tool evidence-retrieval agent

23 therapeutic attributes
GRPO-guided gene-specific reasoning

Gene-adaptive multimodal fusion
Hybrid prior + non-negative PU risk

TargetSage Leads Across 15 Benchmark Tasks

Adjusted F1 = $R_{\text{soft}}^2 / \bar{p}$ $\uparrow$

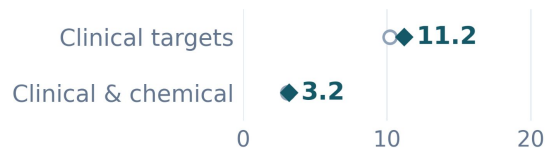
R_{soft} : mean score on known targets
 $\bar{p}$: mean score across evaluated genes

○ Best baseline per task
 ◆ TargetSage

Pharos labels

2 tasks

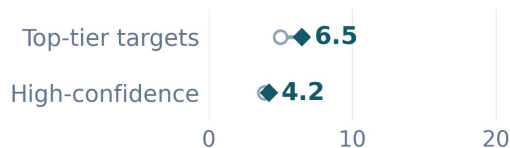
Targets of drugs and active compounds



Druggability tiers

2 tasks

Different levels of supporting evidence



Cancer druggability

6 tasks

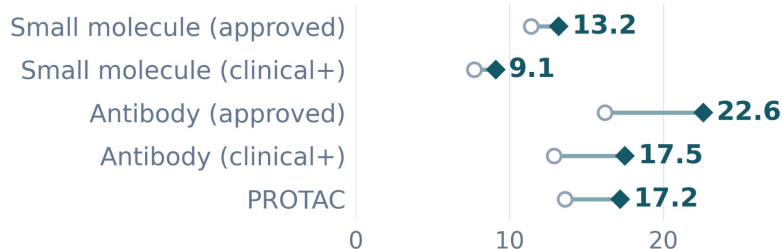
Potential targets for cancer treatment



Drug modality

5 tasks

Small molecules, antibodies and protein degraders



TargetSage Predicts Future Validated Targets

Temporal label holdout and independent clinical-target enrichment

Temporal label holdout

2021

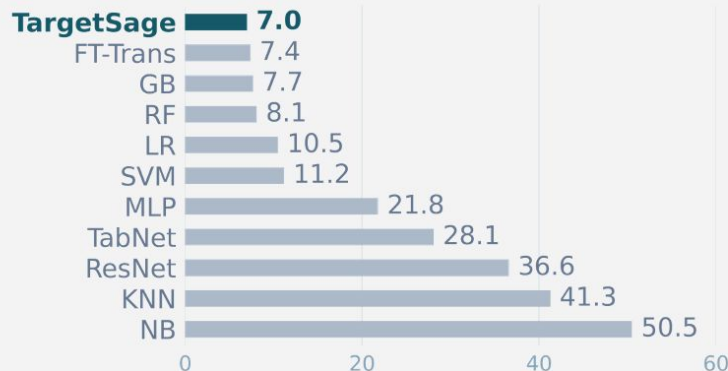
Training labels



2025

90 newly clinical targets

Median rank percentile (%) ↓



Lowest median rank percentile

Independent clinical targets

PHAROS 2025 → ChEMBL Phase II/III

136 clinical targets among 11,891 Tbio genes

proteins targeted by
drugs in phase II or
III clinical trials.

Top-25 enrichment odds ratio ↑



+51% vs. RF (43.2 vs. 28.5)

04

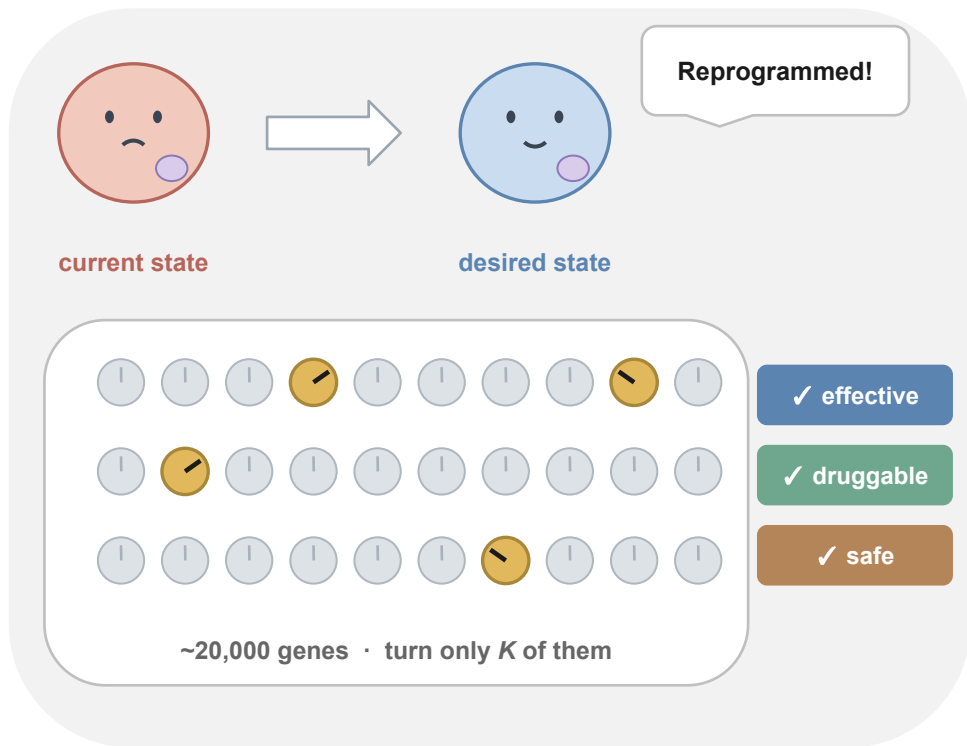
DESIGNING INTERVENTIONS

CellReprogrammer

ICLR 2027 · under review

Which gene combinations shift cells toward a desired state?

Genes interact, so the best few must be chosen together, not one at a time.



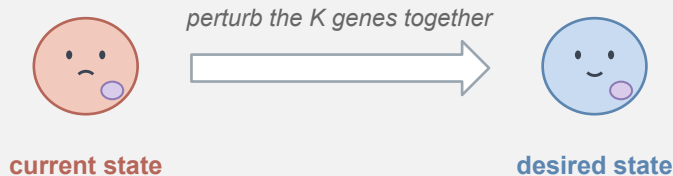
What Is Cell-State Reprogramming?

Perturb a few genes so that cells move toward a desired state

The task



3,000 candidate genes



The desired state depends on the goal:

disease → healthy

this work

healthy → disease

cancer → death

What makes a good gene set?



Effective

Cells really move toward the desired state



Feasible

Only K genes, and existing drugs can move them



Low toxicity

Spare essential genes;
Keep other genes in their normal range

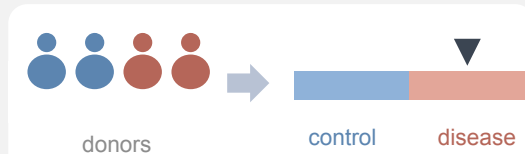
Hard for computers too: "3,000 choose K" possible gene sets, and the effects of genes do not simply add up.

CellReprogrammer: Optimizing Gene Combinations

Balance predicted effectiveness, druggability, and toxicity.

1 Disease Classifier

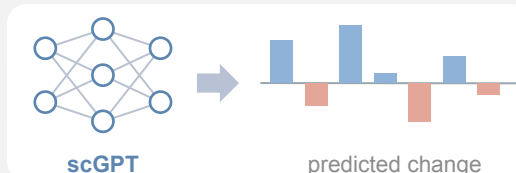
How disease-like is this cell state?



Learn a disease score from donor-level expression profiles of patients and controls.

2 Perturbation Surrogate

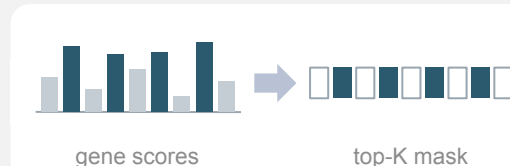
What if we turn genes up or down?



Fine-tune scGPT on Perturb-seq data to predict gene expression responses to candidate perturbations.

3 Combination Optimizer

Which K genes should we pick?



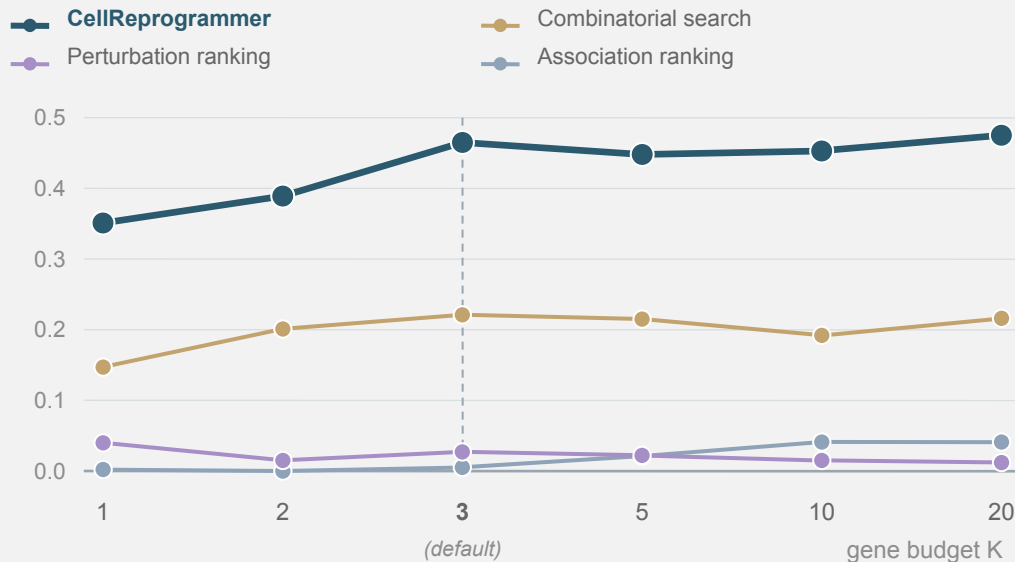
Update gene scores by gradient descent through the predictor and classifier, using differentiable top-K selection.

$$\begin{aligned}
 &\textbf{Objective} \quad \min \quad L(S) \quad + \quad \lambda_{\text{drug}} L_{\text{drug}}(S) \quad + \quad \lambda_{\text{tox}} L_{\text{tox}}(S) \quad \text{s.t.} \quad |S| = K \\
 &\quad \text{Disease score after} \quad \text{Penalty for unsupported} \quad \text{Penalty for essential-gene} \quad \text{K-gene constraint; bounded} \\
 &\quad \text{perturbation} \quad \text{drug directions} \quad \text{inhibition} \quad \text{expression changes}
 \end{aligned}$$

Strongest Predicted Reprogramming at Every Budget K

5 disorders $\times$ 3 cell types $\cdot$ 20 baselines $\cdot$ held-out donors

Effect across gene budgets K



y: predicted strength (higher = stronger) $\cdot$ baseline lines average the methods in each group

At the default budget K = 3

CellReprogrammer

0.47

Best combinatorial search

0.34

Best perturbation ranking

0.28

Best association ranking

0.05

+38%

vs. the best baseline

15 / 15

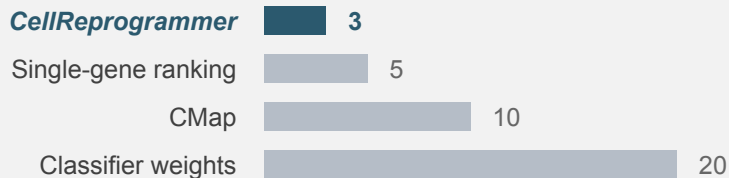
settings moved toward control

Compact, Druggable, and Lower Predicted Toxicity

At the default budget $K = 3 \cdot 5 \text{ disorders} \times 3 \text{ cell types}$

FEASIBILITY

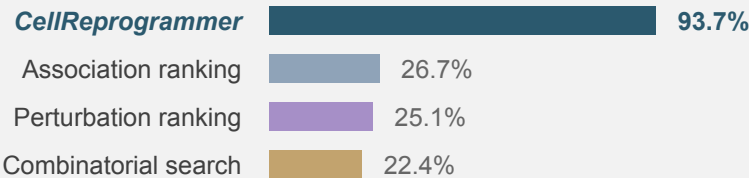
Genes needed for the same effect



to reach strength 0.30 · 10 of 20 baselines never do ($K \leq 20$)

FEASIBILITY

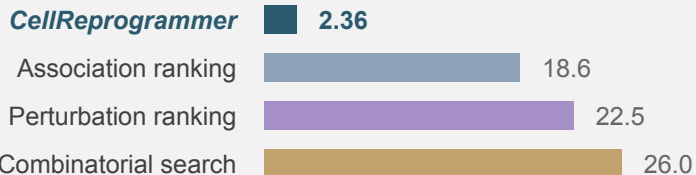
Druggable targets



drug support in the needed direction · category means

TOXICITY

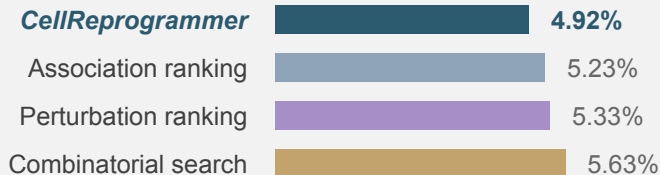
Target-risk score



lower is safer · essential-gene risk · category means

TOXICITY

Downstream genes out of range



lower is better · outside the normal range · category means

Takeaways: From Data to Interventions

Each project answers one question along the way

Decoding disease



Designing interventions

MUSE

CIKM 2025

What tissue patterns are shared across slices?

Builds one tissue map from many slices, even noisy ones.

KEY RESULT

Clearer tissue domains

accurate cross-slice alignment

iMiracle

CIKM 2024

How do neighboring cells influence gene activity?

Decodes which cell–cell signals control which genes.

KEY RESULT

Accurate LR–gene scores

and region-specific signals

TargetSage

NeurIPS 2026

Which genes could be therapeutic targets?

Scores every gene as a target, with reasons a biologist can check.

KEY RESULT

Leads across 15 tasks

predicts future targets

CellReprogrammer

ICLR 2027 · under review

Which gene combinations shift cells toward a desired state?

Picks a few genes that are effective, druggable, and low in toxicity.

KEY RESULT

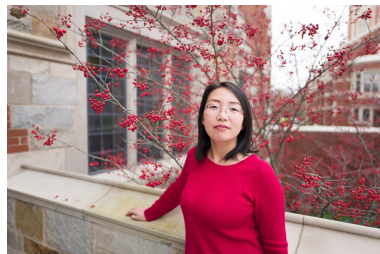
Strongest at every K

vs. 20 baselines · 5 disorders

Together, they turn biological data into testable hypotheses for new treatments.

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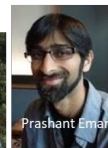
Jennifer Chiarella



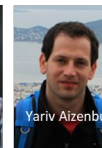
Jay Stanley



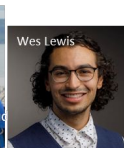
Jimmy Yeh



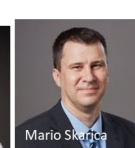
Prashant Emami



Yariv Aizenbud



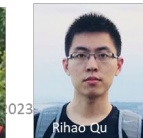
Wes Lewis



Mario Skurica



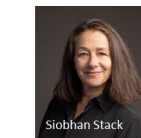
Junchen Yang



Rihao Qu



Boris Landa



Siobhan Stack



Andrew Miao